Lakshya JEE 2.0 2025

Maths

DPP: 5

Matrices

Q1 If $A=\begin{bmatrix}2x & 0\\ x & x\end{bmatrix}$ and $A^{-1}=\begin{bmatrix}1 & 0\\ -1 & 2\end{bmatrix}$, then x equals to:

(A) $1/2$
(B) 1
(C) $-1/2$
(D) 2

Q2 If $A=\begin{bmatrix}6 & 8 & 5\\ 4 & 2 & 3\\ 9 & 7 & 1\end{bmatrix}$ is the sum of a symmetric matrix B and skew -symmetric matrix C, then B is equal to

(A) $\begin{bmatrix}6 & 6 & 7\\ 6 & 2 & 5\\ 7 & 5 & 1\end{bmatrix}$
(B) $\begin{bmatrix}0 & 2 & -2\\ -2 & 5 & -2\\ 2 & 2 & 0\end{bmatrix}$
(C) $\begin{bmatrix}6 & 6 & 7\\ -6 & 2 & -5\\ -7 & 5 & 1\end{bmatrix}$
(D) $\begin{bmatrix}0 & 6 & -2\\ 2 & 0 & -2\\ -2 & -2 & 0\end{bmatrix}$

Q3 The inverse of the matrix $A=\begin{bmatrix}2 & 3 & -3\\ -1 & -2 & 2\\ 1 & 1 & -1\end{bmatrix}$ is:

(A) $\begin{bmatrix}-1 & 2 & 3\\ 1 & 0 & 1\\ -1 & 2 & 1\end{bmatrix}$
(B) $\begin{bmatrix}-5 & -4 & 3\\ -2 & 1 & 3\\ 0 & 1 & 4\end{bmatrix}$
(C) $\begin{bmatrix}-6 & -2 & 1\\ 0 & 1 & 2\\ 2 & -1 & -3\end{bmatrix}$
(D) Does not exist

Q4 Let $A=\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & 2\\ 2 & 2 & 1\end{bmatrix}$, then

(A) $A^2-3A-5I_3=0$
(B) $A^{-1}=\frac{1}{5}(A-4I_3)$
(C) A^3 is not invertible
(D) A^2 is not invertible

Q5 If $A=\begin{pmatrix}\cos\alpha & \sin\alpha\\ -\sin\alpha & \cos\alpha\end{pmatrix}$ and $A^{-1}=A'$, find value of α.

Q6 Let $A=\begin{bmatrix}-1 & 2 & -3\\ -2 & 0 & 3\\ 3 & -3 & 1\end{bmatrix}$ be a matrix, then $|\mathrm{A}|\,\mathrm{adj}\left(\mathrm{A}^{-1}\right)$ is equal to

(A) $\mathrm{O}_{3\times 3}$
(B) $\begin{bmatrix}-1 & 2 & -3\\ -2 & 0 & 3\\ 3 & -3 & 1\end{bmatrix}$
(C) I_3
(D) $\begin{bmatrix}-3 & -3 & 1\\ 3 & 0 & -2\\ -1 & 2 & -3\end{bmatrix}$

Q7 Find the matrix A satisfying the matrix equation:

$$\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} A \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Q8 If A and B are non-singular square matrix of same order 3×3, then which of the following options is correct?

(A) $|\operatorname{adj}(\mathrm{AB})| = |\mathrm{A}||\mathrm{B}|$
(B) $\left|(\operatorname{adj} \mathrm{AB})^{-1}\right| = |\operatorname{adj}(\mathrm{AB})|$
(C) $\left|(\operatorname{adj} \mathrm{AB})^{-1}\right| = \left|\operatorname{adj}(\mathrm{AB})^{-1}\right|$
(D) $\left|\operatorname{adj}(\mathrm{AB})^{\mathrm{T}}\right| = |\mathrm{AB}|^{-2}$

Q9 If A and B are two square matrices such that $\mathrm{B} = -\mathrm{A}^{-1}\mathrm{BA}$, then $(\mathrm{A}+\mathrm{B})^2$ is equal to

(A) O
(B) $\mathrm{A}^2 + \mathrm{B}^2$
(C) $\mathrm{A}^2 + 2\mathrm{AB} + \mathrm{B}^2$
(D) $\mathrm{A} + \mathrm{B}$

Q10 If B is a non-singular matrix and A is a square matrix, then $\det\left(\mathrm{B}^{-1}\mathrm{AB}\right)$ is equal to

(A) $\det\left(\mathrm{A}^{-1}\right)$
(B) $\det\left(\mathrm{B}^{-1}\right)$
(C) $\det(\mathrm{A})$
(D) $\det(\mathrm{B})$

Q11 If

$$B_0 = \begin{bmatrix} -4 & -3 & -3 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{bmatrix}, B_n = \operatorname{adj}(B_{n-1}),$$

$\forall n \in N$

and I is an identity matrix of order 3, then $B_1 + B_3 + B_5 + B_7 + B_9$ is equal to

(A) B_0
(B) $5B_0$
(C) $25B_0$
(D) $5I$

Q12 If $\mathrm{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix}, 6\,\mathrm{A}^{-1} = \mathrm{A}^2 + c\,\mathrm{A}, + \mathrm{dI}$

then $(c, d) =$

(A) $(-6, 11)$
(B) $(-11, 6)$
(C) $(11, 6)$
(D) $(6, 11)$

Q13 Let $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$ and $10B = \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & \alpha \\ 1 & -2 & 3 \end{bmatrix}$.

If B is the inverse of matrix A, then α is

(A) 2
(B) -2
(C) 3
(D) 5

Q14 If $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$, then the matrix A^{-50} when $\theta = \frac{\pi}{12}$, is equal to

(A) $\begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$
(B) $\begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$
(C) $\begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$
(D) $\begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$

Q15 Let M be a 3×3 non-singular matrix with $\det(M) = \alpha$. If M^{-1} adj $(\operatorname{adj} M) = kI$, then the value of k is

(A) 1
(B) α
(C) α^2
(D) α^3

Q16 If the matrix $A = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & -1 \\ 1 & 0 & 2 \end{bmatrix}$, th then compute

$(AB)^{-1}$.

Q17 Find A^{-1} if $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. Also show that $A^{-1} = \frac{A^2 - 3I}{2}$

Q18 Let $A = [a_{ij}]_{3\times 3}$ be a matrix such that $\det(A) = 2$ and $n = \det\left(2\,\mathrm{adj}\left(5A^{-1}\right)\right)$, then number of positive divisors of n is

(A) 8 (B) 10

(C) 12 (D) 14

Answer Key

Q1 (A)

Q2 (A)

Q3 (D)

Q4 (B)

Q5 $\alpha \in \mathrm{R}$

Q6 (B)

Q7 $\mathrm{A} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

Q8 (C)

Q9 (B)

Q10 (C)

Q11 (B)

Q12 (A)

Q13 (D)

Q14 (C)

Q15 (B)

Q16 $= \frac{1}{19} \begin{bmatrix} 16 & 12 & 1 \\ 21 & 11 & -7 \\ 10 & -2 & 3 \end{bmatrix}$

Q17 proof

Q18 (D)

Android App | iOS App | PW Website